Gamaliel Mendoza-Cuevas

PhD. in Neuroscience and quantitative researcher delivering enterprise-grade Data Science solutions.

Executive Summary

Professional Overview

A PhD in Neuroscience with a proven track record of translating complex biological systems into actionable, data-driven insights. My career trajectory reflects a strategic pivot toward Computer and Data Science, positioning me at the forefront of innovation.
As a subject-matter expert, I have built and operationalized end-to-end workflows — spanning database architecture, data-driven analytics, and explainable AI — to deliver measurable, stakeholder-ready outcomes through robust visualizations and executive-level reporting.

Core Competencies & Professional Certifications 👉🔗🖱️

Data Science & ML

Data Science · Machine Learning · Deep Learning · Neural Networks · Computer Vision (CNNs) · NLP · Transformers · LLM Fine-Tuning · Tokenization (BPE) · Explainable AI · Predictive Modeling · Recommender Systems · Clustering (K-Means)

Data Engineering

Data Engineering · ETL Pipelines · Data Warehousing · Data Architecture · Data Quality · MLflow · Model Deployment · Model Serving

Analytics & Research

Data Analytics · Statistical Modeling · Behavioral Analytics · Signal Processing · Research Design · Scientific Writing · R&D Leadership

Languages & Tools

Python · SQL · R · MATLAB · Spark · Databricks · Hadoop · Oracle · MySQL · PyTorch · TensorFlow · Scikit-learn · Git · GitHub · Bitbucket

Professional / Soft Skills

Remote Work · Cross-functional Collaboration · Executive Reporting · Data Visualization · Stakeholder Communication · Bilingual (EN/ES) · German (A2)

Domain

Neuroscience · Data Analytics · Translational Research · Signal Processing

Professional Experience

Cotiviti

Aug 2025 – Aug 2026 | Remote, Mexico City, Mexico.

Associate Data Scientist — Healthcare Analytics & AI Solutions.

- Deployment of model-serving endpoints for neural network classifiers, enabling real-time, scalable healthcare data intelligence.
- Architected and operationalized end-to-end recommendation systems, driving measurable improvements in healthcare service delivery outcomes.
- Engineered K-means clustering-based recommendation engines to optimize resource allocation and elevate stakeholder value.
- Delivered high-performance CNN-based classification solutions, enhancing predictive accuracy across mission-critical healthcare data assets.
- Designed and implemented custom Byte-Pair Encoding (BPE) tokenization pipelines, optimizing downstream NLP performance on enterprise healthcare datasets.
- Led the fine-tuning of Transformer-based language models, unlocking actionable insights from complex, unstructured healthcare data.
- Built and scaled robust ETL pipelines across local and cloud (Databricks) environments, ensuring enterprise-grade data integrity and availability.
- Implemented proactive data quality monitoring frameworks (MLflow), safeguarding model performance and regulatory compliance.
- Delivered a tiered Bronze/Silver/Gold data architecture, establishing a single source of truth to accelerate organization-wide analytics.
- Developed and deployed Python-based, ML-powered applications, translating advanced analytics into scalable, business-ready healthcare solutions.

BIOINVERT

Jan 2017 – Jun 2017 | Mexico City, Mexico

Data Analyst.

Drove standardization and participated in peer-reviewed publication of neuro- and physiological values.

- Conducted advanced CT imaging analysis to quantify bone density, supporting evidence-based clinical decision-making.

- Delivered statistical modeling of age-related bone density trends, generating actionable insights for longitudinal health research.
- Co-authored a peer-reviewed publication accepted in Veterinary Radiology and Ultrasound, reinforcing thought leadership in the field.

APREXBIO

Jan 2016 – Jan 2017 | Mexico City, Mexico

Research Intern.

Led standardization protocols and peer-reviewed publication of neurophysiological benchmark data.

- Executed analysis of visual evoked potentials.
- Achieved certification in MRI equipment operation and acquisition/analysis of anatomical imaging data.
- Co-authored a book chapter establishing standardized benchmark values for visual evoked potentials.
- Co-authored a peer-reviewed publication accepted in Experimental Gerontology.

Academic Credentials 👉🔗🖱️

PhD

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2020 – Apr 2026 | Queretaro, Mexico

Neuroscience.

Doctoral research portfolio spanning translational neuroscience, animal-model methodology, neuropathology, behavioral analytics, and neurophysiology, delivered within a top-tier Biomedical Sciences program.

- Delivered a peer-reviewed publication accepted in the Journal of Neurophysiology.
- Doctoral Thesis: Engineered a low-dose L-DOPA intervention strategy that measurably enhanced visual discrimination performance in a Parkinson's Disease model.
- Doctoral Advisor: Dr. Luis Alberto Carrillo Reid.
- Awarded a competitive doctoral scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation in recognition of research excellence.

Master

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2018 – May 2020 | Queretaro, Mexico

Neurobiology.

Led behavioral assessment of visual discrimination performance within a translational Parkinson's Disease model.

- Graduated with Honorific Mention and a 9.16/10.00 GPA, placing in the top tier of the program cohort.
- Awarded a competitive Master's scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation.

Language Proficiencies 👉🔗🖱️

English

C1 — Professional Working Proficiency (TOEFL iBT: 110/120).

German

A2 — Elementary Proficiency, actively progressing (A1 certified via Goethe-Zertifikat).

Spanish

Native / Bilingual Proficiency.

Selected Publications

Impaired visually guided behavior in a mouse model of Parkinson's disease

Mendoza-Cuevas, G., Perez-Becerra, J., Velazquez-Contreras, R., Calderon, V., Carrillo-Reid, L.

Journal of Neurophysiology, 2025 | DOI: [10.1152/jn.00307.2025](https://doi.org/10.1152/jn.00307.2025)

Visual evoked potentials in the Rhesus monkey

Mendoza-Cuevas, G., Hernández-Godínez, B., Ibáñez-Contreras, A.

Book chapter, 2017 | ISBN: 9781536110753

Computed tomography is a feasible method for quantifying bone density in Macaca mulatta

Solís-Chávez, S., Castillo-Rivera, M., Arteaga-Silva, M., Ibáñez-Contreras, A., Hernández-Godínez, B., Morón-Mendoza, A., Mendoza-Cuevas, G., Morales-Guadarrama, A., Sacristan-Rock, E.

Veterinary Radiology and Ultrasound, 2018 | DOI: [10.1111/vru.12624](https://doi.org/10.1111/vru.12624)

Electrical activity of sensory pathways in female and male geriatric Rhesus monkeys (Macaca mulatta), and its relation to oxidative stress

Ibáñez-Contreras, A., Hernández-Arciga, U., Poblano, A., Arteaga-Silva, M., Hernández-Godínez, B., Mendoza-Cuevas, G., Toledo-Pérez, R., Alarcón-Aguilar, A., González-Puertos, V., Konigsberg, M.

Experimental Gerontology, 2018 | DOI: [10.1016/j.exger.2017.11.003](https://doi.org/10.1016/j.exger.2017.11.003)

The effect of oxidative stress on brain electrical activity and its repercussions on sensory organization in geriatric Rhesus monkeys in captivity

Ibáñez-Contreras, A., Hernández-Godínez, B., Mendoza-Cuevas, G., Hernández-Arciga, U., Königsberg, M.

Book chapter, 2017 | ISBN: 9781536110753

Awards & Recognition

Second Place — 9th State Health Research Forum

Recognized for excellence in visual discrimination research within a Parkinson's Disease model • Querétaro, Mexico • 2024

Third Place — Latin American Brain Initiative (LATBrain) Design Competition

2021